

Suboptimal Decoding

For all block codes used on a memoryless classical channel the optimal decision rule is the *Maximum-likelihood* decoding. In general, the optimal decision rule on classical-quantum channels is not known.

Holevo has introduced the following suboptimal decoding rule $\mathcal{D}_{\mathcal{H}} = (\mathbf{D}_1, \mathbf{D}_2, \dots, \mathbf{D}_M)$ for any block code $\mathcal{B}$ used on c-q channels $\mathcal{C}$ with arbitrary signal states S_i . The decision operators are given by

$$\mathbf{D}_m = \left(\sum_{j=1}^M \mathbf{S}_j^r \right)^{-\frac{1}{2}} \mathbf{S}_m^r \left(\sum_{j=1}^M \mathbf{S}_j^r \right)^{-\frac{1}{2}},$$

$m = 1, 2, \dots, M$, and $r \in (0, 1]$. This decision rule produces a word error probability P_{em} upperbounded by

$$P_{\text{em}} \leq \tilde{P}_{\text{em}} = \sum_{\substack{j=1 \\ j \neq m}}^M \text{Tr} \mathbf{S}_m^{1-r} \mathbf{S}_j^r.$$

Union Bound

Lemma 1 The word error probability P_{em} of the binary code $[N, R]$ used on the binary classical-quantum channel $\mathcal{C}$ and decoded by the suboptimal decision rule $\mathcal{D}_{\mathcal{H}}$ with $r = \frac{1}{2}$ is upperbounded by

$$P_{\text{em}} \leq \sum_{\substack{j=1 \\ j \neq m}}^M 2^{-N[-\underline{d}_{\text{H}}(\mathbf{x}_m, \mathbf{x}_j) \log_2(c)]},$$

where the channel parameter $c := \text{Tr} \sqrt{S_1} \sqrt{S_2}$ and $\underline{d}_{\text{H}}(\mathbf{x}_m, \mathbf{x}_j) = d_{\text{H}}(\mathbf{x}_m, \mathbf{x}_j)/N$ is the normalized Hamming distance between the code words $\mathbf{x}_m$ and $\mathbf{x}_j$.

Consequently, the overall block decoding error probability is upperbounded by the *union bound* given by

$$P_{\text{e}} \leq \tilde{P}_{\text{e}} = \frac{1}{M} \sum_{m=1}^M \sum_{\substack{j=1 \\ j \neq m}}^M 2^{-N[-\underline{d}_{\text{H}}(\mathbf{x}_m, \mathbf{x}_j) \log_2(c)]}.$$

Rescaled Binomial Average Multiplicity Enumerators

Lemma 2 If the average multiplicity enumerators of the codes in the fixed rate sequence $\text{FRS}(R)$ are given by rescaled binomial coefficients

$$\left\{ \underline{M}_n^{(i)} = \frac{a_i(n)}{2^{N_i(1-R)}} \binom{N_i}{n} \right\}_{n=0}^{N_i}, \quad \sum_{n=0}^{N_i} \underline{M}_n^{(i)} = M_i$$

where $a_i(n) = o(2^n)$, then the corresponding IAAME function is given by

$$\mathcal{M}(\underline{\delta}_H, R)_{\text{FRS}} = H_2(\underline{\delta}_H) - (1 - R),$$

where $H_2(x) = -x \log_2(x) - (1 - x) \log_2(1 - x)$ represents the binary entropy function.

Cutoff Rate Lower Bound

Theorem 1 (Cutoff rate lower bound) Fixed rate sequences with rescaled binomial average multiplicity enumerators used on the binary c-q channel $\mathcal{C}$ and decoded by the suboptimal rule $\mathcal{D}_{\mathcal{H}}$ with $r = \frac{1}{2}$ have an error exponent whose lower bound is the cutoff rate bound given by

$$\underline{E}_{\text{cut}}(R) = R_0 - R$$

where R_0 is the cutoff rate given by

$$R_0 = 1 - \log_2(1 + c) .$$

Only code words on the effective Hamming distance

$$\underline{\delta}_{\text{Heff}} = \frac{c}{1 + c}$$

determine the cutoff rate bound. Note that this distance does not depend on the code rate.

Gilbert-Varshamov Bound

The *Gilbert-Varshamov lower bound* on the asymptotic minimal normalized distance of binary block codes $[N, R]$ is given by

$$\underline{\delta}_{\text{GV}}(R) = H_2^{-1}(1 - R),$$

where H_2^{-1} denotes the inverse of the binary entropy function in the interval $[0, \frac{1}{2}]$.

The IAAME of a $\text{FSR}(R)$ with rescaled binomial average multiplicity enumerators satisfying the Gilbert-Varshamov bound is given by the *expurgated IAAME*

$$\mathcal{M}(\underline{\delta}_{\text{H}}, R)_{\text{FSR}} = \begin{cases} -\infty & \text{for } \underline{\delta}_{\text{H}} \leq \underline{\delta}_{\text{GV}}, \\ H_2(\underline{\delta}_{\text{H}}) - (1 - R) & \text{otherwise.} \end{cases}$$

Lemma 3 Fixed rate sequences of linear block codes with rescaled binomial multiplicity enumerators satisfy the Gilbert-Varshamov bound.

Expurgated Lower Bound

Theorem 2 (Expurgated lower bound) Fixed rate sequences with rescaled binomial multiplicity enumerators satisfying the Gilbert-Varshamov bound used on the binary c-q channel $\mathcal{C}$ and decoded by the suboptimal rule $\mathcal{D}_{\mathcal{H}}$ with $r = \frac{1}{2}$ have an error exponent whose lower bound is the expurgated bound given by

$$\underline{E}_{\text{ex}}(R) = \log_2 \left(\frac{1}{c} \right) \cdot \underline{\delta}_{\text{GV}}(R)$$

for rates R below the expurgated rate R_{ex}

$$R_{\text{ex}} = 1 - H_2(\underline{\delta}_{\text{Heff}})$$

and the cutoff rate bound for rates above R_{ex} .

Conclusion

A conceptually simple method for derivation of lower bounds on the error exponent of specific families of block codes used on classical-quantum channels with arbitrary signal states over a finite Hilbert space has been presented. It has been shown that families of binary block codes with appropriately rescaled binomial multiplicity enumerators used on binary c-q channels and decoded by the suboptimal decision rule introduced by Holevo attain the expurgated and cutoff rate lower bounds on the error exponent.

For the classical binary symmetric channel it is known that code families with rescaled binomial multiplicity enumerators attain the tight part of the random coding bound and thus the channel capacity. This has been obtained by an improved upperbounding of the decoding error probability assuming the Maximum-likelihood decoding. Unfortunately, this method cannot be applied directly to binary c-q channels since the optimal decision rule for these channels is not known.